RootCause AI Backend Master Prompt - Part 4 (Final)

CODING STANDARDS

• Use async/await everywhere.
• Keep controllers thin.
• Business logic belongs in services.
• Never call Gemini directly from controllers.
• Validate every request.
• Use reusable utility functions.
• Return consistent JSON responses.

FOLDER RESPONSIBILITIES

config/
Database connection, environment configuration.

controllers/
Receive request, call services, send response.

models/
Mongoose schemas.

routes/
Express routes only.

services/
Business logic.

prompt/
Prompt Builder for Gemini.

middleware/
JWT authentication, upload, validation, error handling.

validators/
Express-validator rules.

utils/
Logger, PDF generator, helper functions.

uploads/
Temporary uploaded files.

END-TO-END FLOW

User Login
↓

JWT Authentication
↓

Dashboard

↓

New Analysis

↓

Upload Files

↓

POST /api/analysis/upload

↓

Multer stores files

↓

Extract text

↓

Prompt Builder

↓

Gemini Service

↓

Analysis Service

↓

Store JSON in MongoDB

↓

Return analysisId

↓

Frontend redirects

/app/history/:analysisId

↓

GET /api/analysis/:id

↓

Display Analysis Details

DATABASE RULES

Users can only:
• View their own analyses
• Delete their own analyses
• Export only their own reports

Never expose another user's data.

INTERVIEW READY ARCHITECTURE

Be able to explain:

Why JWT?
Why bcrypt?
Why Multer?
Why MongoDB?
Why Gemini 2.5 Flash?
Why Prompt Builder?
Why Service Layer?
Why structured JSON?
Why HTTP-only cookies?
Why Helmet?
Why Rate Limiting?
Why Validation?

PRODUCTION CHECKLIST

✓ JWT Authentication
✓ MongoDB Atlas
✓ Multer Uploads
✓ Gemini API
✓ Secure Middleware
✓ Validation
✓ Error Handling
✓ Modular Architecture
✓ Dashboard APIs
✓ Analysis CRUD
✓ PDF Export
✓ Environment Variables
✓ Logging
✓ Deployment Ready

FINAL CURSOR AI INSTRUCTION

Generate a production-quality Express backend following this document exactly.

Do not simplify the architecture.

Use reusable components, clean folder structure, modular services, secure coding practices, and production-ready code suitable for deployment and for a resume-quality AI SaaS application.